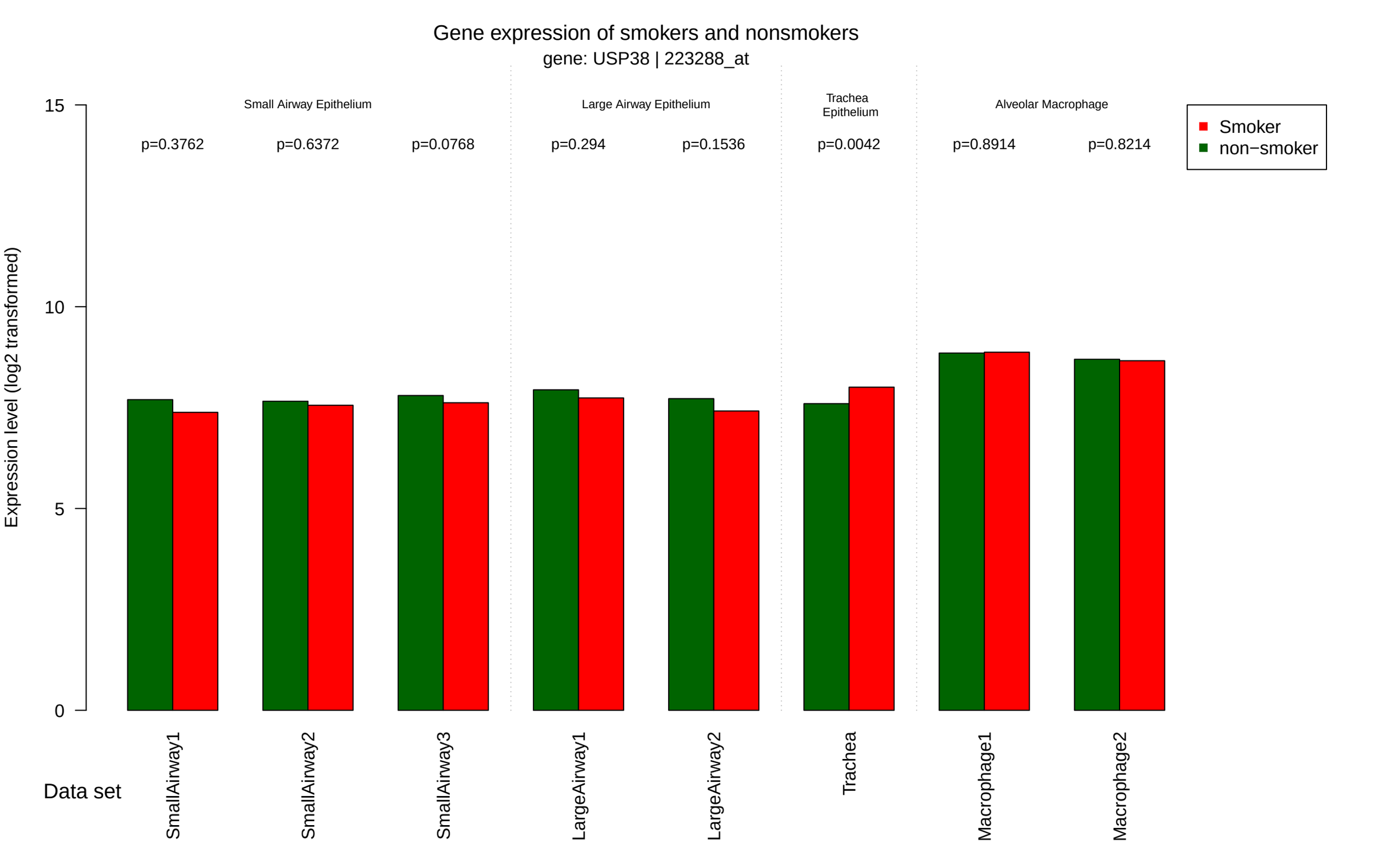
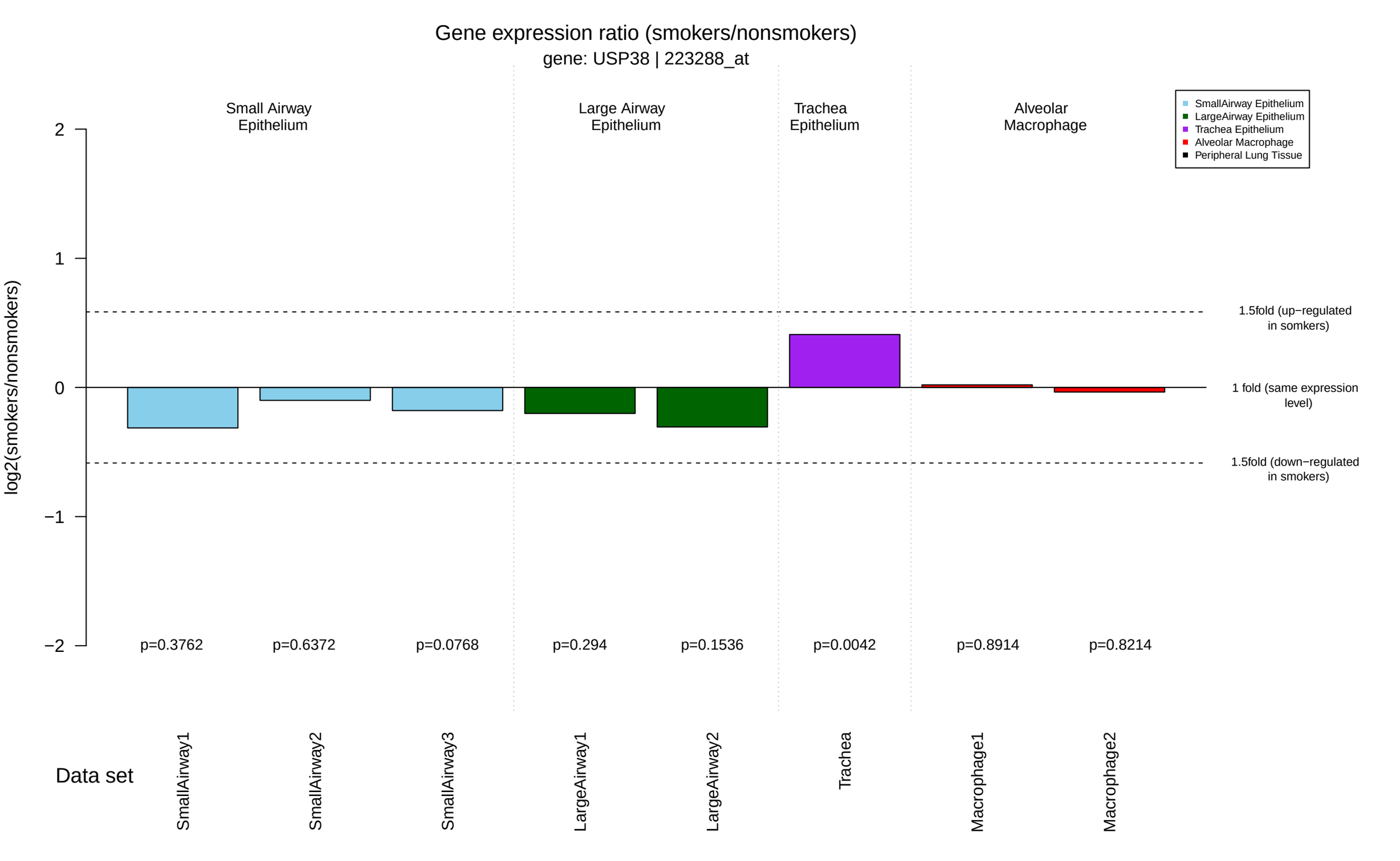
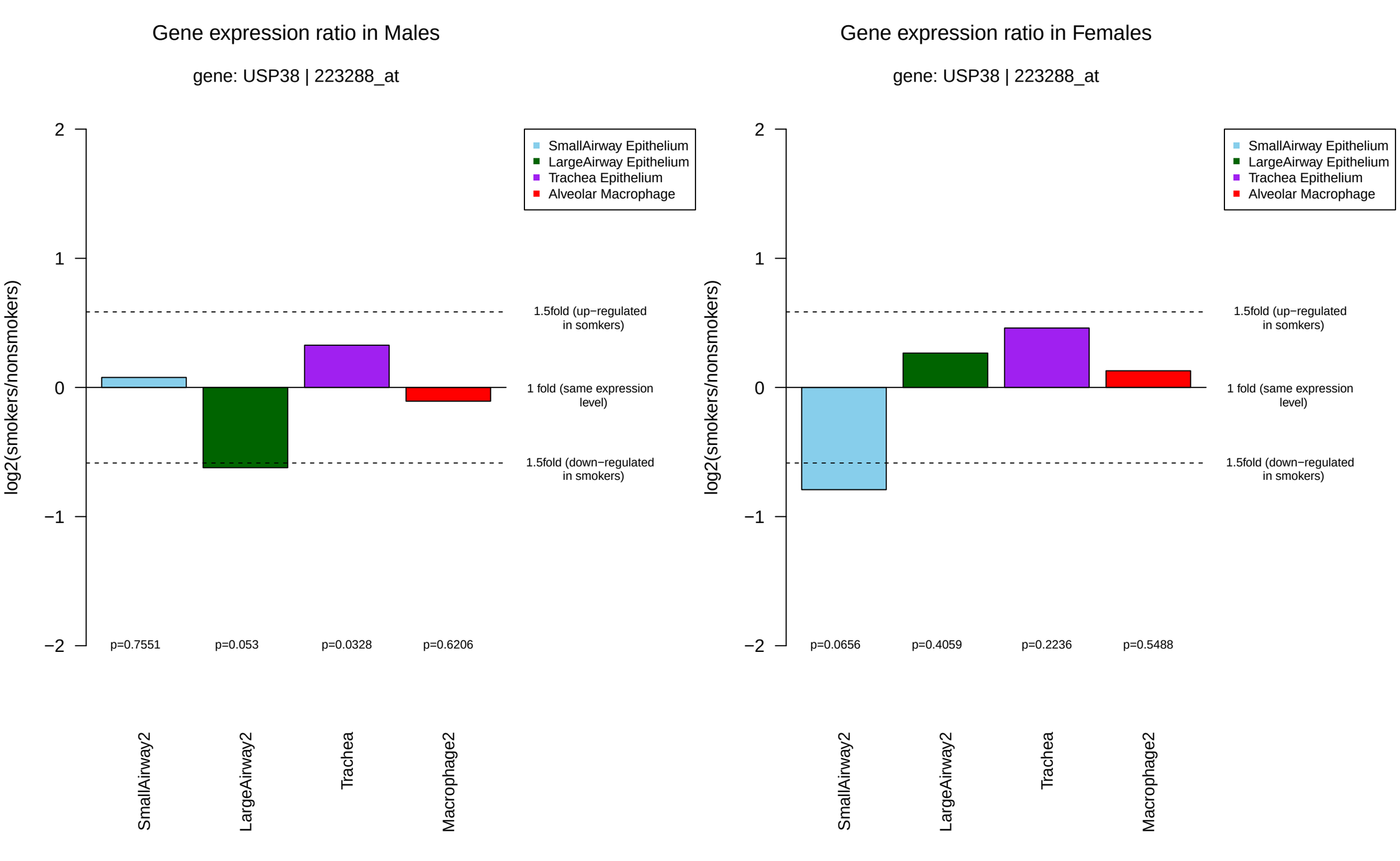
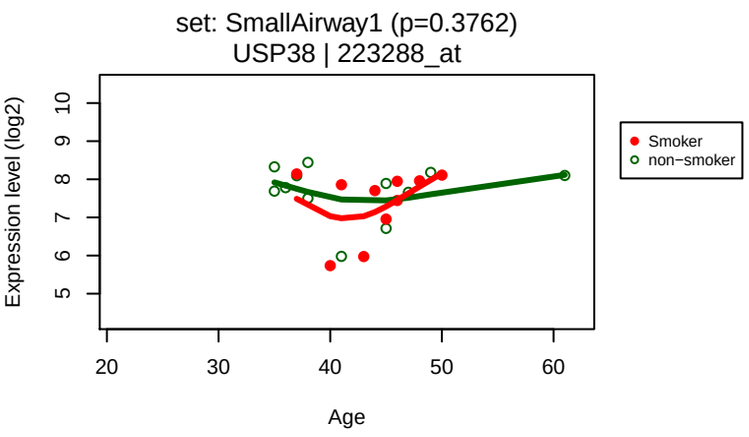
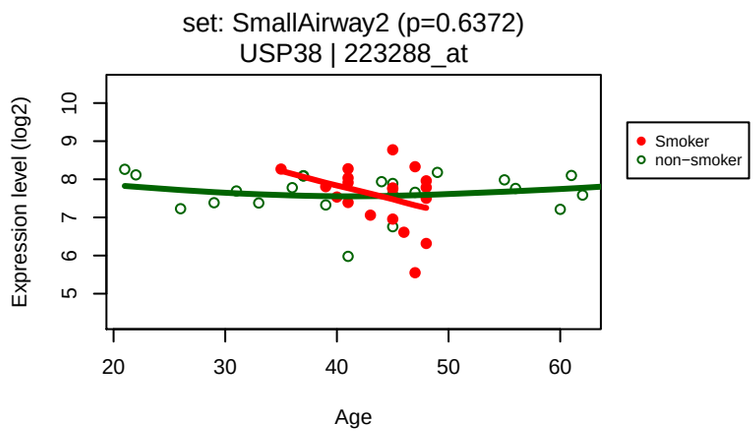
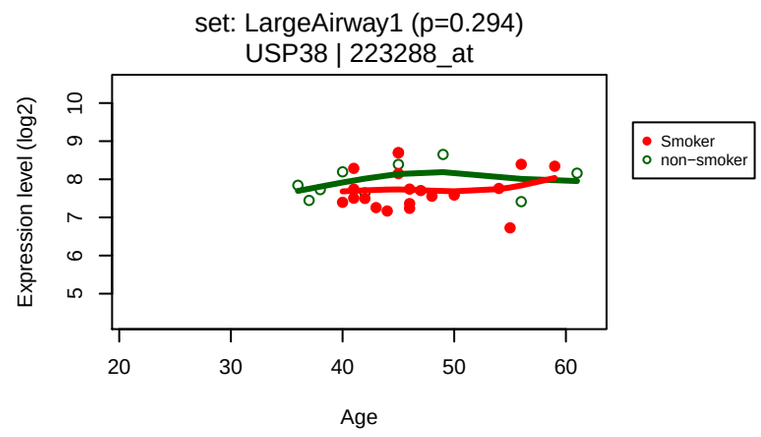

Gene expression of smokers and nonsmokers
gene: USP38 | 223288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3762
p=0.6372
p=0.0768
p=0.294
p=0.1536
p=0.0042
p=0.8914
p=0.8214
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

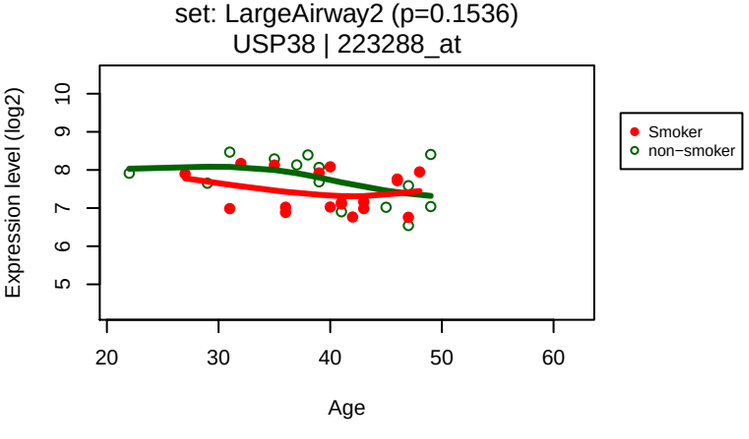

Gene expression ratio (smokers/nonsmokers)
gene: USP38 | 223288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3762
p=0.6372
p=0.0768
p=0.294
p=0.1536
p=0.0042
p=0.8914
p=0.8214
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

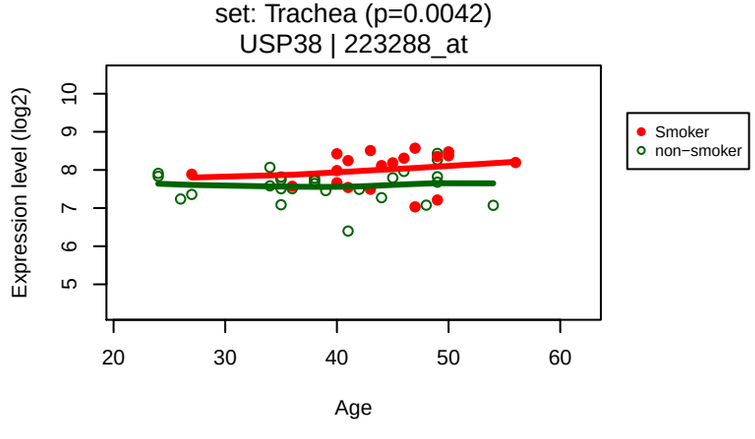

Gene expression ratio in Males
gene: USP38 | 223288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7551
p=0.053
p=0.0328
p=0.6206
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP38 | 223288_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0656
p=0.4059
p=0.2236
p=0.5488
SmallAirway2
LargeAirway2
Trachea
Macrophage2

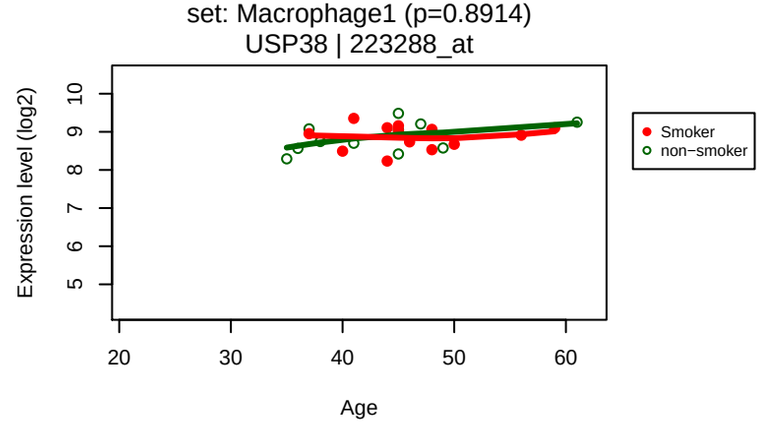

set: SmallAirway1 (p=0.3762)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker

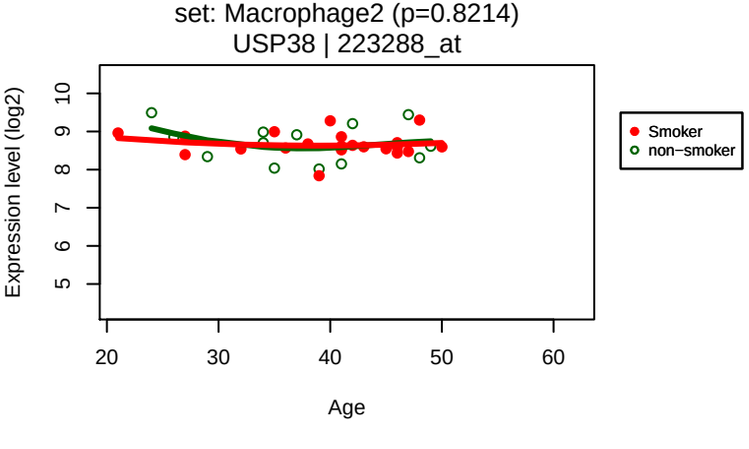

set: SmallAirway2 (p=0.6372)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker

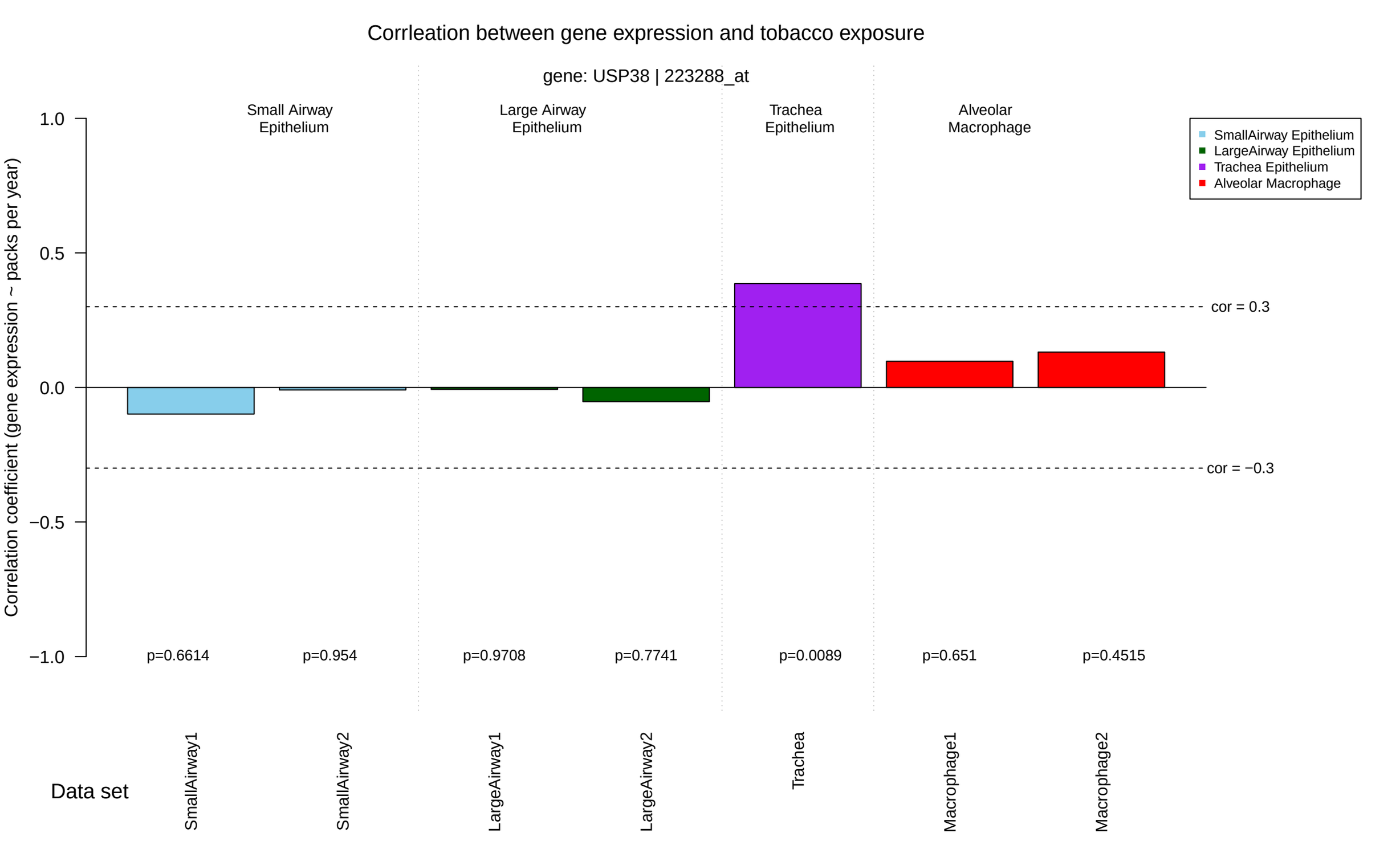

set: LargeAirway1 (p=0.294)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1536)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0042)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8914)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8214)
USP38 | 223288_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP38 | 223288_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6614
p=0.954
p=0.9708
p=0.7741
p=0.0089
p=0.651
p=0.4515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2